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Measure Theory

1.1 Outer Measure

Definition 1.1.1 ► Outer Measure

A function φ defined for every subset A of an arbitrary set X is called an **outer measure** on X if the following conditions are satisfied:

- (i) $\varphi(\emptyset) = 0$,
- (ii) $0 \leq \varphi(A) \leq \infty$ whenever $A \subset X$,
- (iii) $\varphi(A_1) \leq \varphi(A_2)$ whenever $A_1 \subset A_2$,
- (iv) $\varphi(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \varphi(A_i)$ for every countable collection of sets $\{A_i\}$ in X .

Example 1.1.2 ► Outer Measure

- (i) In an arbitrary set X , define $\varphi(A) = 1$ if A is nonempty and $\varphi(\emptyset) = 0$.
- (ii) Let $\varphi(A)$ be the number (possibly infinite) of points in A .
- (iii) Let $\varphi(A) = \begin{cases} 0 & \text{if card } A \leq \aleph_0, \\ 1 & \text{if card } A > \aleph_0. \end{cases}$
- (iv) If X is a metric space, fix $\varepsilon > 0$. Let $\varphi(A)$ be the smallest number of balls of radius ε that cover A .
- (v) Select a fixed x_0 in an arbitrary set X , and let

$$\varphi(A) = \begin{cases} 0 & \text{if } x_0 \notin A, \\ 1 & \text{if } x_0 \in A; \end{cases}$$

φ is called the **Dirac measure** concentrated at x_0 .

Definition 1.1.3

Let φ be an outer measure on a set X . A set $E \subset X$ is called **φ -measurable** if

$$\varphi(A) = \varphi(A \cap E) + \varphi(A - E)$$

for every set $A \subset X$. In view of property (iv) above, observe that φ -measurability requires only

$$\varphi(A) \geq \varphi(A \cap E) + \varphi(A - E).$$

Lemma 1.1.4

A set $E \subset X$ is φ -measurable if and only if

$$\varphi(P \cup Q) = \varphi(P) + \varphi(Q)$$

for all sets P and Q such that $P \subset E$ and $Q \subset E^c$.

Proof. For Necessity, if E is φ -measurable, then for all such P, Q ,

$$\varphi(P \cup Q) = \varphi((P \cup Q) \cap E) + \varphi((P \cup Q) - E) = \varphi(P) + \varphi(Q)$$

For sufficiency, let $P = A \cap E$, $Q = A \cap E^c = A \setminus E$. Then

$$\varphi(A) = \varphi(P \cup Q) = \varphi(P) + \varphi(Q) = \varphi(A \cap E) + \varphi(A \setminus E)$$

□

Theorem 1.1.5

Suppose φ is an outer measure on an arbitrary set X . Then the following four statements hold:

- (i) E is φ -measurable whenever $\varphi(E) = 0$.
- (ii) $\emptyset$ and X are φ -measurable.
- (iii) $E_1 - E_2$ is φ -measurable whenever E_1 and E_2 are φ -measurable.
- (iv) If $\{E_i\}$ is a **countable collection of disjoint φ -measurable sets**, then $\bigcup_{i=1}^{\infty} E_i$ is φ -measurable and

$$\varphi\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \varphi(E_i).$$

More generally, if $A \subset X$ is an arbitrary set, then

$$\varphi(A) = \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c),$$

where

$$S = \bigcup_{i=1}^{\infty} E_i.$$

Proof. 1. Take any $A \subseteq X$. Since $A \cap E \subseteq E$, $\varphi(A \cap E) = 0$. Thus

$$\varphi(A \cap E) + \varphi(A \cap E^c) = \varphi(A \cap E^c) \leq \varphi(A)$$

2. Follows immediately from lemma 1.1.4.

3. Take any P, Q such that $P \subseteq (E_1 \setminus E_2) = E_1 \cap E_2^c$, $Q \subseteq (E_1 \setminus E_2)^c = E_1^c \cup E_2$.

$$\varphi(Q) = \varphi(Q \cap E_2) + \varphi(Q - E_2)$$

Note that $Q - E_2 \subseteq E_1^c$, $Q \cap E_2 \subseteq E_2$. Since $P \subseteq E_1$, the measurability of E_1 implies that

$$\varphi(P) + \varphi(Q - E_2) = \varphi(P \cup (Q - E_2))$$

Note that $P \cup (Q - E_2) \subseteq E_2^c$. Since $Q \cap E_2 \subseteq E_2$, we have

$$\varphi(P \cup (Q - E_2)) + \varphi(Q \cap E_2) = \varphi(P \cup Q)$$

Finally, we have

$$\begin{aligned} \varphi(P) + \varphi(Q) &= \varphi(P) + \varphi(Q \cap E_2) + \varphi(Q - E_2) \\ &= \varphi(P \cup (Q - E_2)) + \varphi(Q \cap E_2) \\ &= \varphi(P \cup Q) \end{aligned}$$

4. Let $S_k = \bigcup_{i=1}^k E_i$. We proceed by finite induction and first note that the result is obviously true for $k = 1$. For $k > 1$ assume that S_k is φ -measurable and that

$$\varphi(A) \geq \sum_{i=1}^k \varphi(A \cap E_i) + \varphi(A \cap S_k^c), \quad \forall A \subseteq X$$

Then

$$\begin{aligned}\varphi(A) &= \varphi(A \cap E_{k+1}) + \varphi(A \cap E_{k+1}^c) \\ &= \varphi(A \cap E_{k+1}) + \varphi(A \cap E_{k+1}^c \cap S_k^c) + \varphi(A \cap E_{k+1}^c \cap S_k) \\ &= \varphi(A \cap E_{k+1}) + \varphi(A \cap S_{k+1}^c) + \varphi(A \cap S_k)\end{aligned}$$

Replaced A by $A \cap S_k$, we have

$$\varphi(A \cap S_k) \geq \sum_{i=1}^k \varphi(A \cap S_k \cap E_i) + \varphi(A \cap S_k \cap S_k^c) = \sum_{i=1}^k \varphi(A \cap E_i)$$

Thus

$$\varphi(A) \geq \sum_{i=1}^{k+1} \varphi(A \cap E_{k+1}) + \varphi(A \cap S_{k+1}^c)$$

Then the subadditivity gives that

$$\varphi(A) \geq \varphi(A \cap S_{k+1}) + \varphi(A \cap S_{k+1}^c)$$

this, in turn, implies that S_{k+1} is φ -measurable. Since we now know for every set $A \subseteq X$ and for all positive integers k

$$\varphi(A) \geq \sum_{i=1}^k \varphi(A \cap E_i) + \varphi(A \cap S_k^c)$$

We have

$$\varphi(A) \geq \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c) \geq \varphi(A \cap S) + \varphi(A \cap S^c)$$

where $S = \bigcup_{i=1}^{\infty} E_k$. Thus S is measurable. The subadditivity gives that

$$\varphi(A) \leq \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c)$$

Finally, we have

$$\varphi(A) = \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap S^c)$$

Replace A by S , we can get

$$\varphi(S) = \sum_{i=1}^{\infty} \varphi(E_i)$$

□

Lemma 1.1.6

Let $\{E_i\}$ be a sequence of arbitrary sets. Then there exists a sequence of disjoint sets $\{A_i\}$ such that each $A_i \subset E_i$ and

$$\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} A_i.$$

If each E_i is φ -measurable, and so is A_i .

Proof. Let $S_k = \bigcup_{i=1}^k E_i$. Let $A_k = S_k \setminus S_{k-1}$, $k \geq 1$, where $S_0 = \emptyset$. Then

$$\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} (S_k \setminus S_{k-1}) = \bigcup_{i=1}^{\infty} A_i.$$

If each E_i is φ -measurable. $E_{i+1} \cup E_i = (E_{i+1} - E_i) \cup E_i$ is measurable follows from the (ii), (iii) of the above theorem. Inductively, every $S_k = (S_{k-1} \setminus E_k) \cup E_k$ is measurable. Then each A_k is measurable. □

Theorem 1.1.7

If $\{E_i\}$ is a sequence of φ -**measurable** sets in X , then $\bigcup_{i=1}^{\infty} E_i$ and $\bigcap_{i=1}^{\infty} E_i$ are φ -**measurable**.

Proof. By the lemma above, there exists a sequence of disjoint φ -measurable sets $\{A_i\}$ such that each $A_i \subseteq E_i$, and

$$\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{\infty} A_i$$

It follows from Theorem 1.1.5 that $\bigcup_{i=1}^{\infty} A_i$ is measurable, then $\bigcup_{i=1}^{\infty} E_i$ is as

well . Since X is measurable , $E_i^c = X \setminus E_i$ is measurable , then $\bigcup_{i=1}^{\infty} E_i^c$ is measurable . Thus

$$\bigcap_{i=1}^{\infty} E_i = X \setminus \bigcup_{i=1}^{\infty} E_i^c$$

is measurable . □

Definition 1.1.8

A nonempty collection Σ of sets E satisfying the following two conditions is called a **σ -algebra**:

- (i) if $E \in \Sigma$, then $E^c \in \Sigma$;
- (ii) $\bigcup_{i=1}^{\infty} E_i \in \Sigma$ if each E_i is in Σ .

Definition 1.1.9

In a topological space, the elements of the smallest σ -algebra that contains all open sets are called **Borel sets**. The term “smallest” is taken in the sense of inclusion.

Proof. We need to show that the “smallest” exists. □

Corollary 1.1.10

If φ is an **outer measure** on an arbitrary set X , then the class of φ -measurable sets forms a **σ -algebra**.

Proof. Follows immediately from Theorem 1.1.5 and Theorem 1.1.7. □

Corollary 1.1.11

Suppose φ is an outer measure on X and $\{E_i\}$ a countable collection of φ -measurable sets.

- (i) If $E_1 \subset E_2$ with $\varphi(E_1) < \infty$, then

$$\varphi(E_2 \setminus E_1) = \varphi(E_2) - \varphi(E_1).$$

- (ii) (Countable additivity) If $\{E_i\}$ is a disjoint sequence of sets, then

$$\varphi\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \varphi(E_i).$$

(iii) (Continuity from the left) If $\{E_i\}$ is an increasing sequence of sets, that is, if $E_i \subset E_{i+1}$ for each i , then

$$\varphi\left(\bigcup_{i=1}^{\infty} E_i\right) = \varphi\left(\lim_{i \rightarrow \infty} E_i\right) = \lim_{i \rightarrow \infty} \varphi(E_i).$$

(iv) (Continuity from the right) If $\{E_i\}$ is a decreasing sequence of sets, that is, if $E_i \supset E_{i+1}$ for each i , and if $\varphi(E_{i_0}) < \infty$ for some i_0 , then

$$\varphi\left(\bigcap_{i=1}^{\infty} E_i\right) = \varphi\left(\lim_{i \rightarrow \infty} E_i\right) = \lim_{i \rightarrow \infty} \varphi(E_i).$$

(v) If $\{E_i\}$ is any sequence of φ -measurable sets, then

$$\varphi\left(\liminf_{i \rightarrow \infty} E_i\right) \leq \liminf_{i \rightarrow \infty} \varphi(E_i).$$

(vi) If

$$\varphi\left(\bigcup_{i=i_0}^{\infty} E_i\right) < \infty$$

for some positive integer i_0 , then

$$\varphi\left(\limsup_{i \rightarrow \infty} E_i\right) \geq \limsup_{i \rightarrow \infty} \varphi(E_i).$$